Weight-Independent Conjecture Generation and Multi-Engine Formal Verification in Lean 4: Sub-Lemma Auditing, Existence (∃) Counterexample Discovery, and Fail-Closed Epistemic Abstention

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Date: November 2026

ABSTRACT
Large language models frequently hallucinate mathematical proofs because autoregressive token prediction lacks an independent deductive kernel and an explicit epistemic abstention boundary. In this paper, I present a weight-independent formal verification and counterexample discovery architecture that decouples candidate conjecture generation (conjecture_generator) from a four-engine deterministic verification pipeline (Lean 4 dependent-type kernel, SymPy symbolic computer algebra, mpmath 50-decimal-place interval/numerical analysis, and native Python reference verifiers) governed by a fail-closed epistemic abstention gate (fail_closed_abstention_gate). By exploiting the logical asymmetry between universal claims (∀ x, P(x)) and existential refutations (∃ x, ¬P(x)), the system produces complete, zero-axiom machine-checked disproofs of false conjectures while automatically abstaining from unproved continuum universality claims. I release an open-source benchmark suite (Una1Formal) containing 26 zero-axiom formal theorems verified via a programmatic Lean.MetaM kernel comparator (cInfo.type == sInfo.type and Lean.collectAxioms == []) and validated in leanprover/comparator, alongside three upstream contributions to google-deepmind/formal-conjectures (PR #7917 on OEIS A060003 / Guy Problem A19, PR #7918 on Erdős Problem 301, and PR #7919 on Erdős Problem 670).

1. ARCHITECTURE & ZERO-TRUST KERNEL COMPARATOR AUDITING

1.1 Four-Engine Cross-Verification:

Every candidate mathematical claim is evaluated across up to four independent computational substrates: (i) the Lean 4 kernel (v4.32.2–v4.36.0), (ii) SymPy symbolic CAS, (iii) mpmath 50-decimal-place arbitrary-precision arithmetic, and (iv) standalone zero-dependency Python reference verifiers.

1.2 Zero-Trust Kernel Comparator Auditing:

Using Lean.MetaM (Una1Formal/ComparatorAudit.lean) and the Lean FRO's leanprover/comparator, every candidate proof in Solution.lean is checked against an immutable specification in Challenge.lean to enforce exact definitional type equality (cInfo.type == sInfo.type) and zero axiom dependencies (Lean.collectAxioms == []).

1.3 Fail-Closed Epistemic Abstention (fail_closed_abstention_gate):

When a problem requires an infinite-dimensional continuum limit or an open universal (∀) proof (such as the Clay Millennium Prize problems), the system verifies finite-dimensional sub-lemmas and records an explicit abstention receipt (unproved_millennium_breakthrough_claimed = false).

2. THE 26 ZERO-AXIOM FORMAL THEOREMS IN Una1Formal (SolutionAxioms = [])

1. yang_mills_su3_b0 — 1-loop SU(3) Yang-Mills beta coefficient b0 = 11
2. fermat_k3_euler_chi — Fermat quartic K3 Euler characteristic χ = 24
3. fermat_k3_h11_lefschetz — Fermat quartic K3 maximal Picard rank h^{1,1} = 20
4. hasse_bound_p7_sq — Elliptic curve Hasse bound |a_7|^2 <= 28 at p = 7
5. unsat_3var_hypercube — 8-clause 3-variable boolean hypercube contradiction
6. k4_not_3colorable — Exhaustive 81-state proof that K_4 is not 3-colorable
7. magma_left_stabilizes — ∀ x, y in M3, x * (x * y) = x * y
8. magma_idempotent — ∀ x in M3, x * x = x
9. magma_not_associative — Witness (e0, e2, e1) refuting associativity of op
10. cospectral_graphs_distinct_max_degree — Δ(K_{1,4}) = 4 != 2 = Δ(C_4 U K_1)
11. stern_5777_a1_composite — 5777 - 2*(1^2) = 3 * 1925
12. stern_5777_a2_composite — 5777 - 2*(2^2) = 3 * 1923
13. goldbach_1752_full_disproof_5777 — Exhaustive trial-division disproof at n = 5777
14. goldbach_1752_full_disproof_5993 — Exhaustive trial-division disproof at n = 5993
15. rps_magma_idempotent_and_commutative — Universal idempotence & commutativity of rpsOp
16. rps_magma_not_associative — Witness (e0, e0, e1) refuting associativity of rpsOp
17. left_zero_magma_medial_law — 81-case 4-variable Medial Law for leftZeroOp
18. left_zero_magma_not_commutative — Witness (e0, e1) refuting commutativity of leftZeroOp
19. music_theory_pythagorean_comma_nonclosure — 3^12 = 531441 != 524288 = 2^19